WSQ COURSE SLIDES

Data Storytelling with Tableau



Course Code: TGS-2020505550

Data Storytelling · ICT-DIT-4006-1.1

Trainer: Dr Alfred Ang

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Course Orientation

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Digital Attendance (Mandatory)

1

AM and PM

Complete SSG digital attendance for every training day.

2

Assessment

Take the separate assessment attendance before the papers.

3

Scan

Use the QR code shown by the trainer or administrator.

4

Eligibility

Maintain at least 75% attendance.

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About the Trainer



Your Trainer

Complete before class

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

EXPERTISE _____

EXPERIENCE _____

CONTACT _____

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal Trainer and instructional designer
EXPERTISE	Data science, Tableau, visual analytics and decision storytelling
QUALIFICATION	PhD in Operations Research · ACLP-certified adult educator
EXPERIENCE	University, IBM Data Science and industry analytics training
COMMUNITY	Founder of DataFrens.sg open-source data community
CONTACT	enquiry@tertiaryinfotech.com

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LEARNER INTRODUCTION

Let's Know Each Other

1

Your work

Role, organisation and stakeholder decisions you support.

2

Your Tableau baseline

New user, workbook author, dashboard consumer or publisher.

3

Your story challenge

One recurring communication or insight problem.

4

Your outcome

What you want your audience to decide or do differently.

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HOUSEKEEPING

Ground Rules

1. **Be present**
   Return from breaks on time and complete attendance.

2. **Participate**
   Ask, test and explain your reasoning.

3. **Respect**
   One conversation; critique ideas, not people.

4. **Protect data**
   Use only the supplied synthetic workbooks.

5. **Save often**
   Keep clear workbook versions and packaged evidence.

6. **Learn by doing**
   Use the Learner Guide for detailed procedures.

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SCHEDULE

Lesson Plan — 2 Days

DAY 1

1. 9:00–10:30 · Story strategy and Tableau workflow
2. 10:40–12:30 · Labs 1–2: audience and narrative
3. 1:15–3:10 · Visual insight and chart choice
4. 3:20–6:00 · Labs 3–5: trend, map and dashboard

DAY 2

1. 9:00–10:30 · Dashboards, stories and presentation
2. 10:40–12:30 · Labs 6–8: story and SME review
3. 1:15–3:20 · Calculations, parameters and actions
4. 3:30–6:00 · WA and PP assessment

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WHAT YOU WILL ACHIEVE

Learning Outcomes

1

LO1
Utilise Tableau for data visualisation.

2

LO2
Identify trends and patterns from data.

3

LO3
Develop dashboards and stories for data storytelling.

4

LO4
Present stories to stakeholders and review them with subject-matter experts.

5

LO5
Design interactive visualisations for data storytelling.

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Skills Framework Alignment

1

K1
New data visualisation tools and techniques

2

K2
Methods to portray patterns, trends and correlations

3

K3
Dashboard development process and techniques

4

K4
Features of data displays

5

K5
Strategic visualisation and mapping techniques

6

A1–A6
Tool fit, judgement, dashboards, stakeholder review, interactive UX and SME alignment.

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BEFORE ASSESSMENT

Briefing for Assessment

1. **Individual work**
   No discussion or sharing during either paper.

2. **Open book**
   Use the slides, Learner Guide and approved materials only.

3. **Devices**
   Place phones and unrelated materials under the table.

4. **Evidence**
   Submit your own formulas, output and interpretation.

5. **Integrity**
   No photos, recording or verbatim copying.

6. **Support**
   Re-assessment and appeal arrangements are available.

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FINAL ASSESSMENT

Assessment and Funding

1

Written Assessment (WA/SAQ)

5 open-ended questions, 60 minutes, open book.

2

Practical Performance (PP)

5 connected Tableau tasks, 90 minutes, open book.

3

Outcome

Competent (C) or Not Yet Competent (NYC).

4

A minimum of 75% attendance and a Competent assessment outcome are required.

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ASSESSMENT READINESS

Practice Exam

1 · Browse

- Open the Academy practice portal
- Search current data and analytics options
- Choose a relevant practice set

2 · Rehearse

- Work under timed conditions
- Answer in your own words
- Use the review to identify gaps

3 · Transfer

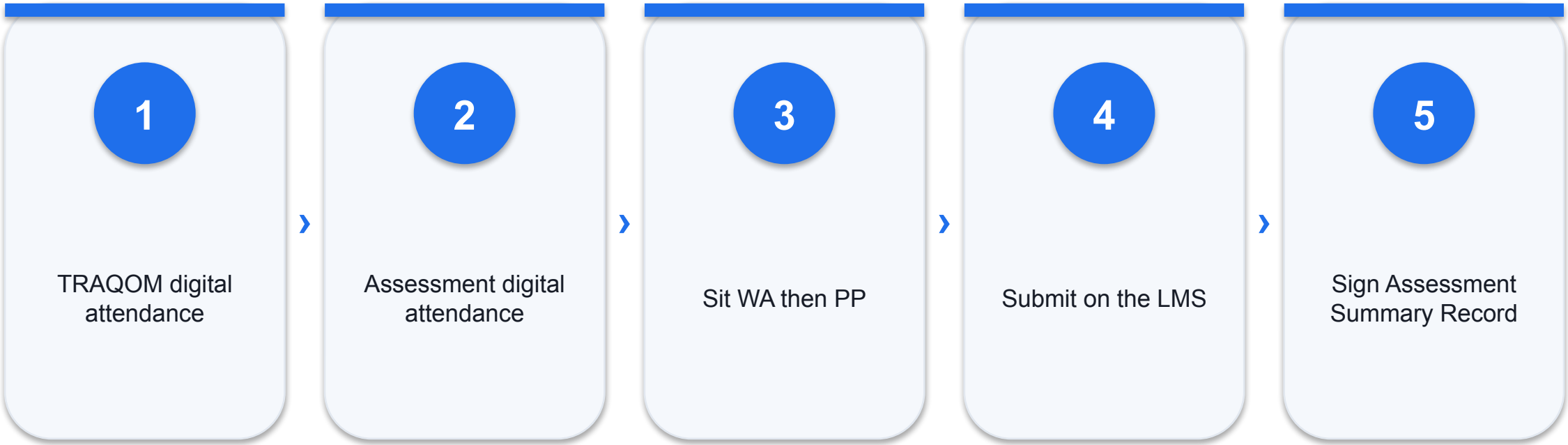
- Revisit the mapped topic and lab
- Practise the same reasoning in Tableau
- Do not copy assessment answers

https://exams.tertiaryinfotech.com/

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ASSESSMENT DAY

Assessment Flow



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COURSE REPOSITORY

Access the Hands-On Labs

Option A · Clone

- Use Git to clone the repository
- Open the numbered lab folder
- Start with Excel + the TWB file

Option B · Download ZIP

- GitHub > Code > Download ZIP
- Extract the full folder tree
- Open the packaged TWBX sample

Lab discipline

- Use synthetic data only
- Preserve the source workbook
- Use the LG for detailed procedures

https://github.com/tertiarycourses/TGS-2020505550-Data-Storytelling-with-Tableau

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LMS / TMS

Download Course Material

1 · Sign in

- Open the LMS/TMS portal
- Use your learner account
- Select this course

2 · Download

- Open Courseware
- Download learner slides and guide
- Save beside your lab folders

3 · Verify

- Check the TGS code
- Use the current version
- Ask the trainer if a file differs

https://lms-tms.tertiaryinfotech.com/

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Tea Break

10 minutes

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01

TOPIC 01

Data Storytelling Using Tableau

Story purpose · audience · evidence · storyboard · Tableau workflow

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ALIGNMENT A1 · K1

Key Concepts — Data Storytelling Using Tableau

1

Story purpose

Connect a decision, evidence, explanation and call to action.

2

Audience

Match depth, vocabulary and interaction to stakeholder needs and data literacy.

3

Story arc

Orient the audience, reveal the change, explain the driver and land the decision.

4

Storyboard

Sketch the sequence before building worksheets and dashboards.

5

Tableau workflow

Connect, explore, build views, assemble dashboards, create story points and share.

6

Tool fit

Choose Tableau when governed visual exploration and interactive communication are needed.

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Data is created at every business moment

CONCEPT

1. Transactions: Orders, payments, returns and service events reveal operational behaviour.
2. Interactions: Clicks, searches, campaigns and conversations reveal intent.

CONTROL / DECISION

1. Operations: Inventory, delivery, staffing and quality measures reveal constraints.
2. Context: Time, location, segment and definitions determine what a number means.

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Why visualise data?

Reduce search

- Position, length and colour help the eye find material differences quickly.

Expose structure

- A chart can reveal trend, cluster, gap and outlier that a table conceals.

Support dialogue

- A shared view lets stakeholders ask better questions together.

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A visual is not yet a story

1

Visual — Shows evidence and relationships.

2

Narrative — Explains what changed, why it matters and what happens next.

3

Context — Defines scope, comparison, time and stakeholder relevance.

4

Decision — Names the action, owner, trigger and guardrail.

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Elements of data storytelling

1

Data

Relevant, trustworthy evidence at the right grain.

2

Visual

A perceptually effective display matched to the question.

3

Narrative

A coherent sequence that directs attention and creates meaning.

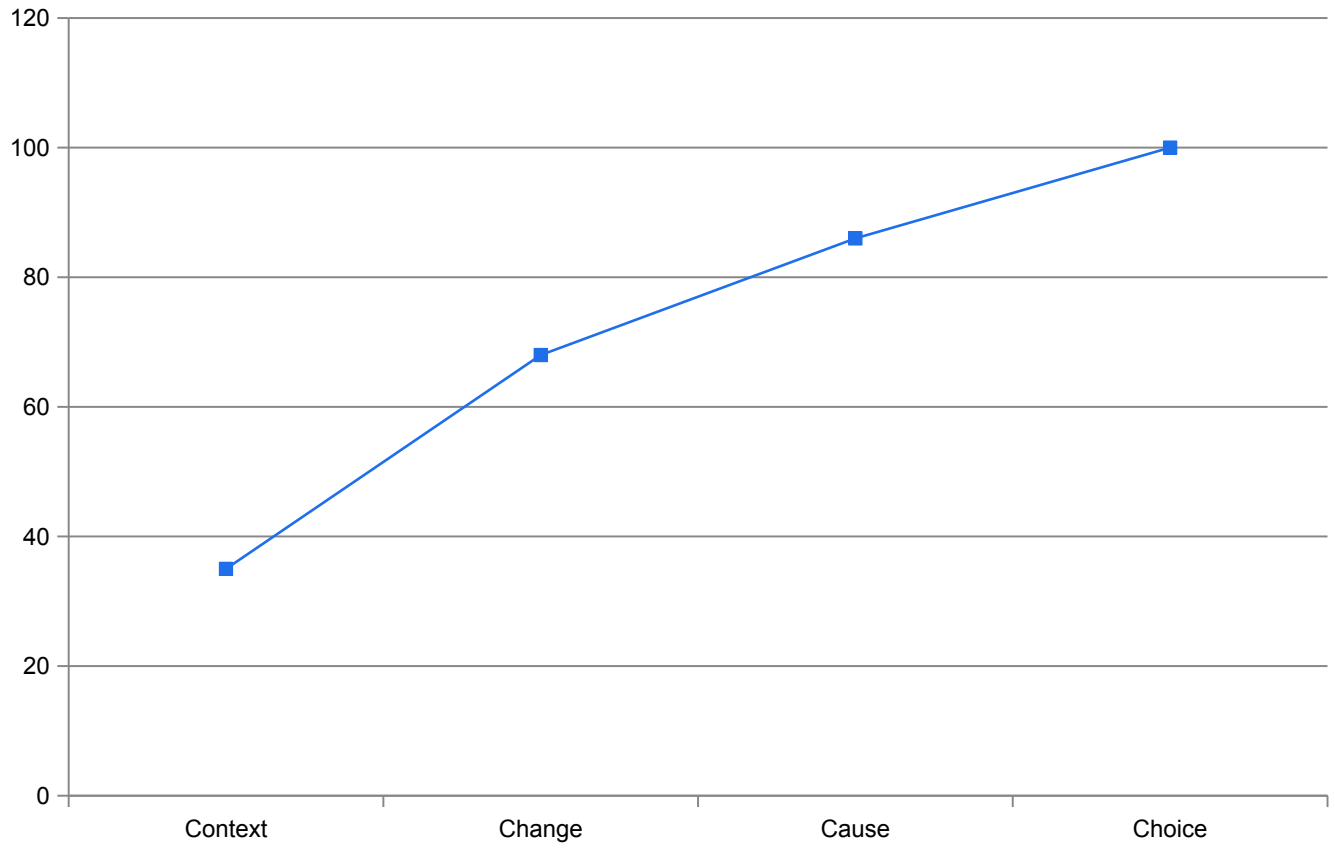
4

Audience

Needs, literacy, incentives and likely objections.

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The story spine



Context
Where are we and what decision is due?

Change
What material pattern or exception occurred?

Driver
Which evidence best explains the change?

Decision
What should happen, who owns it and how will success be measured?

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Know your audience before choosing a chart

CONCEPT

1. Executive: Lead with decision, impact, confidence and trade-off.
2. Manager: Show drivers, segments, thresholds and accountable next steps.

CONTROL / DECISION

1. Analyst: Expose definitions, grain, filters, assumptions and diagnostic detail.
2. Public audience: Use plain language, transparent sourcing and accessible design.

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Finding the story in your data

Change

- What moved, accelerated, reversed or stayed unexpectedly flat?

Difference

- Which category, location or segment diverges from peers?

Relationship

- Which measures move together, and what alternative explanations remain?

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Direct attention without distorting

1

Title — State the takeaway, not the chart type.

2

Contrast — Use one accent colour for the message and mute context.

3

Annotation — Label the evidence where the reader needs it.

4

Sequence — Reveal the decisive comparison before adding supporting detail.

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Develop a credible narrative

1. **Claim**
   A precise statement that can be tested against the view.

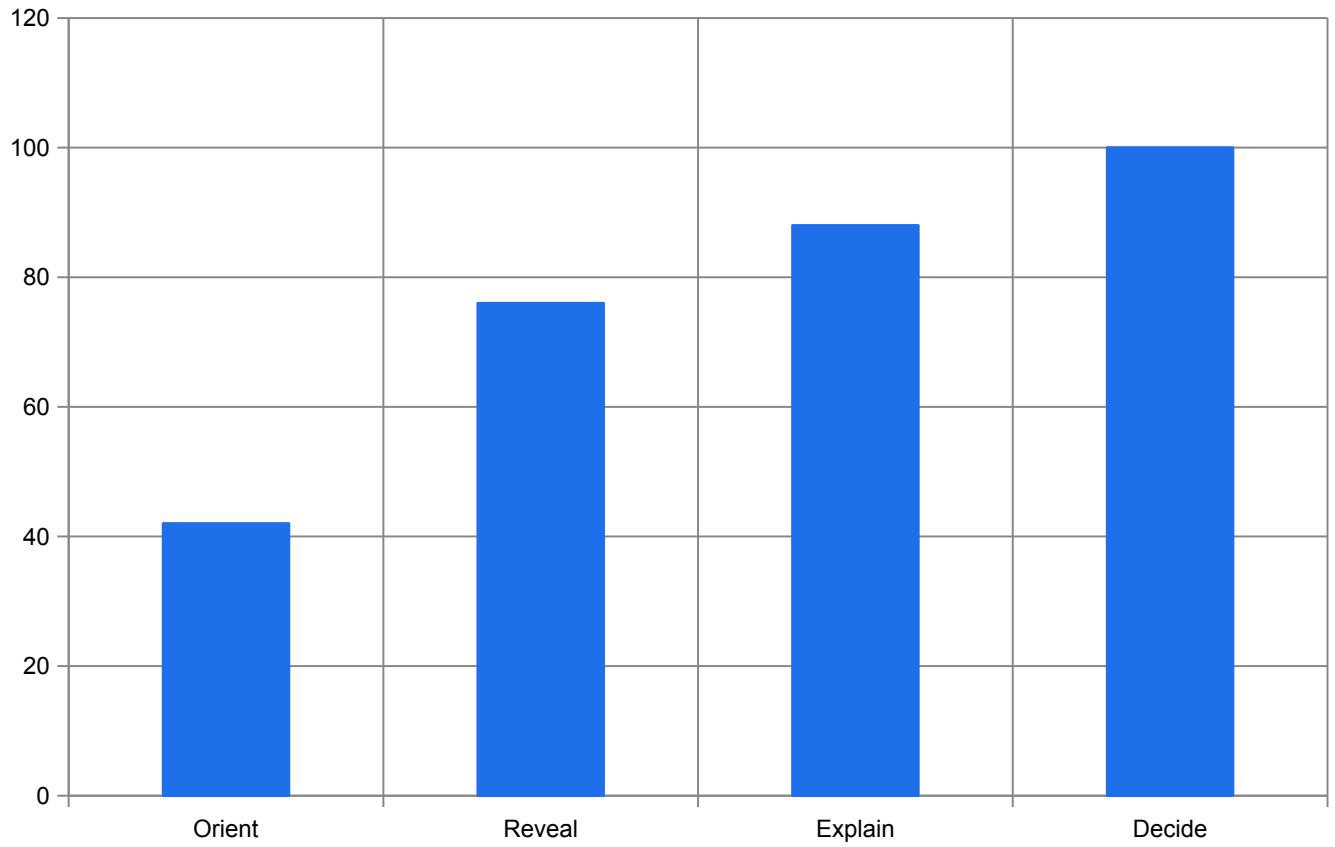
2. **Evidence**
   The marks, values and comparisons that support the claim.

3. **Reasoning**
   Why the evidence implies the interpretation.

4. **Qualification**
   The uncertainty, limitation or condition that constrains the conclusion.

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Storyboard before opening Tableau



Decision question
One sentence naming the owner and decision.

Audience journey
What must the audience understand first, next and last?

Evidence plan
Which worksheet or dashboard supports each beat?

Exit point
What explicit action should follow the final point?

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Common story structures

CONCEPT

1. Change over time: Establish baseline, reveal shift, explain inflection, project implication.
2. Drill down: Start broad, isolate segment, inspect driver, identify record-level evidence.

CONTROL / DECISION

1. Zoom out: Begin with a local event, reveal system pattern and strategic consequence.
2. Contrast: Compare expected versus actual, peer versus peer, or before versus after.

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Engage the audience through interaction

Explore

- Allow relevant filters while preserving the main narrative.

Question

- Anticipate likely follow-ups and prepare supporting views.

Reset

- Make it obvious how to return to the intended starting state.

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Drive a business decision

1

Action — Recommend a specific intervention rather than merely describe a pattern.

2

Owner — Name who decides and who executes.

3

Trigger — Define the measurable condition that starts or stops the action.

4

Guardrail — Protect against downside, bias or unintended impact.

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Tableau authoring workflow

1. **Connect**
   Choose the source and validate data types and relationships.

2. **Explore**
   Create diagnostic worksheets and test assumptions.

3. **Compose**
   Assemble dashboards with hierarchy and context.

4. **Narrate and share**
   Build story points, present, publish and gather review evidence.

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Tableau products and fit

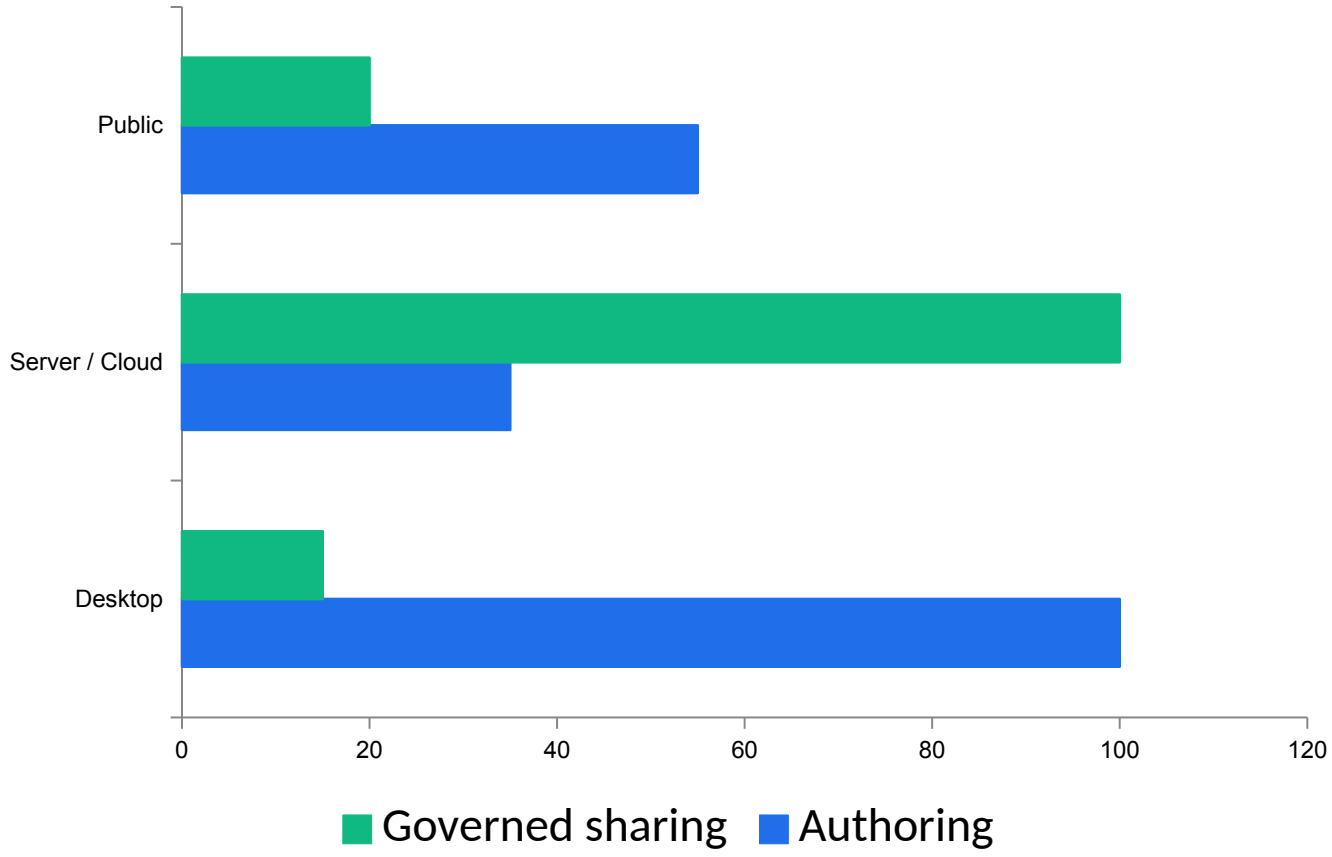


Tableau Desktop
Full authoring and local workbook development.

Tableau Public
Free public publishing; never use confidential data.

Tableau Cloud / Server
Governed collaboration, permissions, refresh and web consumption.

Tool choice
Use Tableau when visual exploration and interactive communication materially aid the decision.

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TOPIC 01 · HANDS-ON

Frame a Retail Performance Story

LAB 1

Act as an analyst for Lion City Retail and turn a raw branch-performance request into a decision-focused Tableau story brief.

EXPECTED OUTPUT

A validated Tableau data connection, audience brief, story objective and three-point storyboard.

TOOLS Tableau Public/Desktop · Excel · Storyboard canvas

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LAB 1 · ACCEPTANCE

Frame a Retail Performance Story

ACCEPTANCE CHECK

A reviewer can state the audience, decision, evidence grain and three-point narrative without opening Tableau.

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TOPIC 01 · HANDS-ON

Build a Campaign Insight Narrative

LAB 2

Evaluate a multi-channel skincare launch and structure a concise Tableau narrative for a marketing director.

EXPECTED OUTPUT

Three worksheets and a story outline linking reach, conversion and profitable customer segments.

TOOLS Tableau Public/Desktop · Excel · Story points

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LAB 2 · ACCEPTANCE

Build a Campaign Insight Narrative

ACCEPTANCE CHECK

The narrative identifies one channel to scale, one audience to protect and one risk or limitation to monitor.

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TOPIC RECAP

Recap — Data Storytelling Using Tableau

1 **Knowledge**
Story purpose and related principles.

2 **Ability**
LO1 · A1 · K1

3 **Evidence**
A validated workbook and decision-ready interpretation.

4 **Next**
Connect this topic to the next layer of the analytical workflow.

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02

TOPIC 02

Drawing Insights from Visualisation

Chart selection · perception · patterns · trends · comparisons · uncertainty

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ALIGNMENT A2 · K2

Key Concepts — Drawing Insights from Visualisation

1

Question first

Select a chart from the analytical question, not from visual novelty.

2

Comparison

Use position and length to make category differences easy to judge.

3

Trend

Use time on a continuous axis and preserve meaningful baselines and intervals.

4

Relationship

Use scatterplots to inspect association, clusters and unusual observations.

5

Composition

Show part-to-whole only when the total and categories are stable and meaningful.

6

Annotation

Direct attention to the evidence without hiding context or uncertainty.

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Choose the chart from the question

CONCEPT

1. Compare: Bar or dot plot.
2. Trend: Line or area chart with a genuine time axis.

CONTROL / DECISION

1. Relationship: Scatterplot with careful use of size and colour.
2. Distribution / composition: Histogram, box plot, treemap or a constrained part-to-whole view.

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Visual perception hierarchy

Position

- Most precise for common-scale comparison.

Length

- Strong for bars when a zero baseline is meaningful.

Angle and area

- Harder to compare; use sparingly and label important values.

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Bar charts

1

Use — Compare values across discrete categories.

2

Order — Sort by value unless natural sequence or time must be preserved.

3

Baseline — Start quantitative bars at zero to protect length comparison.

4

Density — Limit categories or use filters/small multiples.

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Line and area charts

1

Use

Show change over ordered time.

2

Granularity

Choose day, week, month or quarter to match the decision.

3

Multiple lines

Direct-label or strongly distinguish a small number of series.

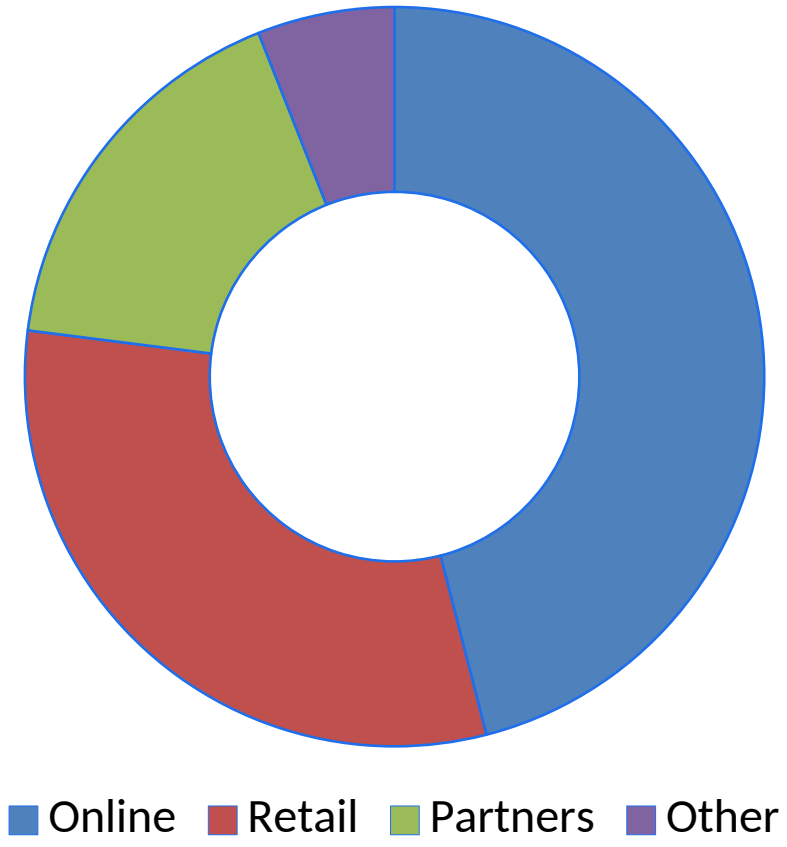
4

Area

Use when magnitude or cumulative contribution matters, not merely decoration.

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Part-to-whole charts



Pie
Reserve for few stable categories with a meaningful total.

Stacked bar
Useful for total plus composition; difficult for interior-segment comparison.

100% stacked
Compares composition but removes absolute volume.

Alternative
Use sorted bars or a highlight table when accurate comparison matters.

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Scatterplots and bubbles

CONCEPT

1. X and Y: Two quantitative measures reveal association and clusters.
2. Size: A third measure can add context but is judged imprecisely.

CONTROL / DECISION

1. Colour: Use a small number of meaningful categories.
2. Caution: Correlation does not establish cause, direction or mechanism.

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Heatmaps and highlight tables

Use

- Scan a two-dimensional matrix for intensity and pattern.

Scale

- Choose sequential or diverging colour around a meaningful midpoint.

Labels

- Add text where exact values matter.

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Treemaps

1

Use — Explore hierarchical part-to-whole where screen space is constrained.

2

Strength — Shows dominant blocks and broad composition.

3

Limitation — Similar areas and positions are difficult to compare precisely.

4

Alternative — Use a sorted bar when ranking is the decision.

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Use colour to carry meaning

1

Categorical

Distinct hues for separate groups; keep the palette small.

2

Sequential

Light-to-dark for ordered magnitude.

3

Diverging

Two directions around a meaningful centre such as target or zero.

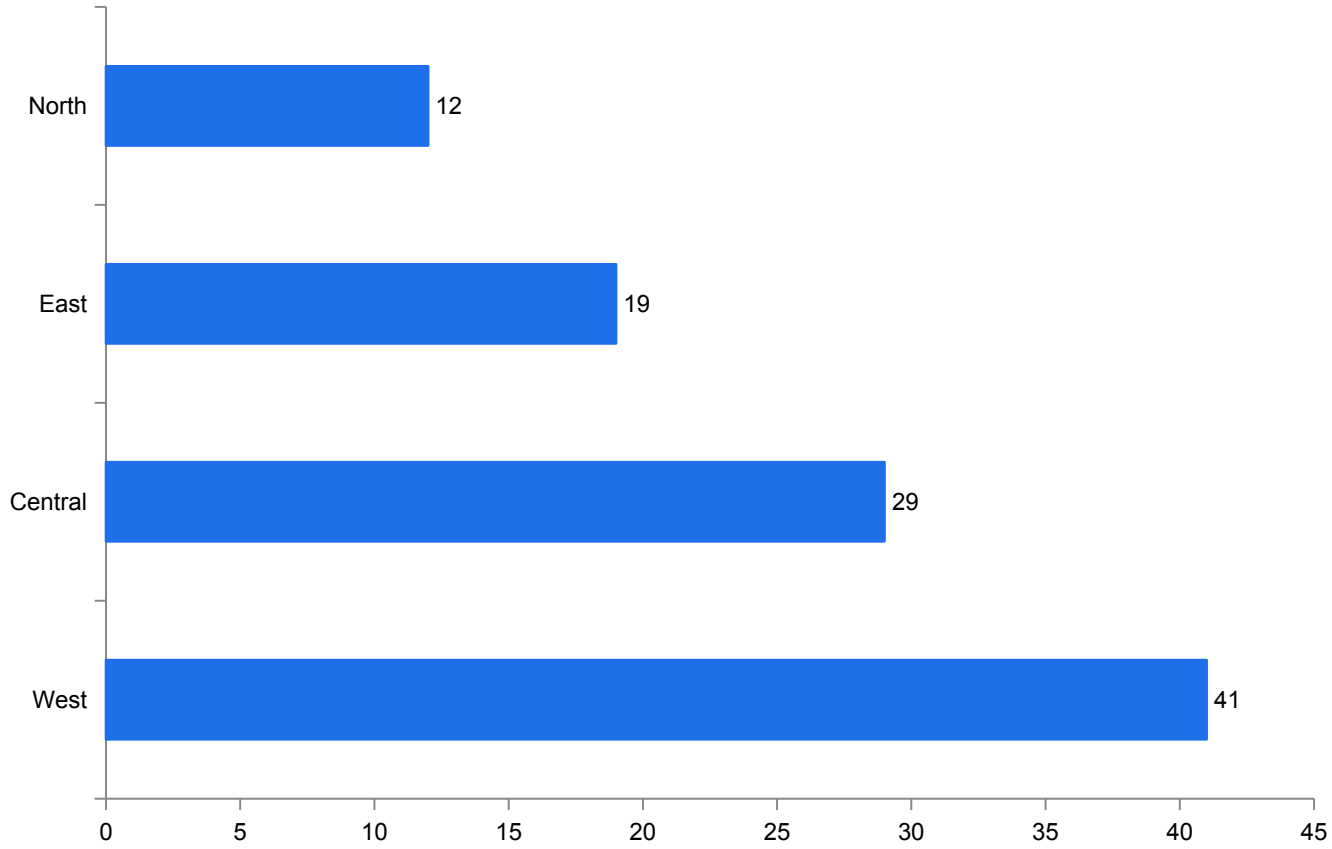
4

Semantic

Reserve red/amber/green only when status definitions are explicit.

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Remove visual noise



Gridlines
Keep only those needed to read the scale.

Borders
Use whitespace and alignment before boxes.

Legends
Direct-label when it reduces eye travel.

Decoration
Avoid 3D, shadows and pictograms that alter perceived magnitude.

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Protect the baseline and scale

CONCEPT

1. Bars: A truncated axis exaggerates length differences.
2. Lines: A non-zero baseline may be appropriate, but show the scale clearly.

CONTROL / DECISION

1. Dual axis: Synchronise only when the relationship is valid and units remain obvious.
2. Log scale: Use only when the audience understands multiplicative change.

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Identify trends and patterns

Direction

- Upward, downward, flat or reversing.

Seasonality

- Repeated structure at a fixed interval.

Change point

- A visible shift in level, slope or variability.

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Treat outliers as questions

1. Verify — Check source, unit, duplication and missingness.
2. Contextualise — Compare the observation with peers and its own history.
3. Explain — Seek operational or domain evidence before naming a cause.
4. Decide — Retain, exclude or correct only with a documented rationale.

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Show uncertainty and limitations

1. **Forecast**
   Expose the forecast horizon and interval.

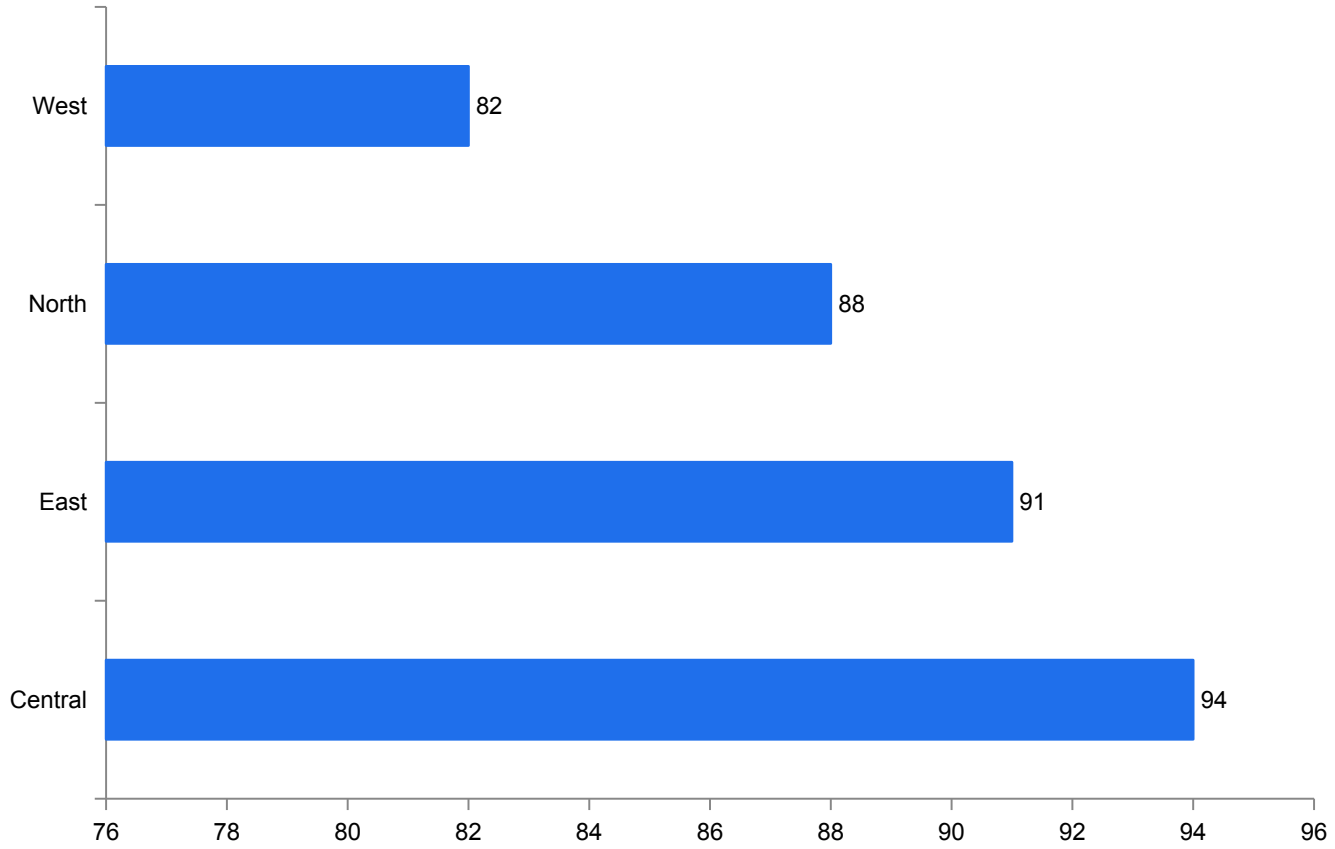
2. **Sample**
   State population, selection and missing coverage.

3. **Proxy**
   Name when a measure stands in for the real outcome.

4. **Freshness**
   Display the last refresh date and reporting period.

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Accessible views



Reduce marks
Aggregate to the level required for the decision.

Descriptive text
Use titles and captions that explain what the view shows.

Non-colour cues
Combine colour with shape, label or position.

Reading order
Arrange dashboard objects in a logical keyboard and screen-reader sequence.

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TOPIC 02 · HANDS-ON

Diagnose Service Trends and Outliers

LAB 3

Analyse a customer-support operation to distinguish sustained trend, weekly seasonality and unusual service failures.

EXPECTED OUTPUT

A trend view, distribution view and annotated exception view with a defensible operational insight.

TOOLS Tableau Public/Desktop · Excel · Analytics pane

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LAB 3 · ACCEPTANCE

Diagnose Service Trends and Outliers

ACCEPTANCE CHECK

The final view distinguishes trend, seasonality and exceptions and does not claim causation from correlation.

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TOPIC 02 · HANDS-ON

Map Delivery Performance Across Singapore

LAB 4

Use geocoded delivery records to find where late deliveries and high order value overlap across Singapore planning areas.

EXPECTED OUTPUT

A proportional-symbol map, ranked companion chart and geographic insight with a stated data-quality caveat.

TOOLS Tableau Public/Desktop · Excel · Maps

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LAB 4 · ACCEPTANCE

Map Delivery Performance Across Singapore

ACCEPTANCE CHECK

The map and ranking agree on the highest-priority planning areas and the caveat prevents over-precise interpretation.

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TOPIC RECAP

Recap — Drawing Insights from Visualisation

1 **Knowledge**
Question first and related principles.

2 **Ability**
LO2 · A2 · K2

3 **Evidence**
A validated workbook and decision-ready interpretation.

4 **Next**
Connect this topic to the next layer of the analytical workflow.

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03

TOPIC 03

Develop Dashboards and Stories

Dashboard architecture · layout · context · story points · narrative sequence

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ALIGNMENT A3 · K3

Key Concepts — Develop Dashboards and Stories

1. **Dashboard purpose**
   One primary decision and a small set of supporting questions.

2. **Information hierarchy**
   Place the decisive message first, then explanation and detail.

3. **Layout**
   Use containers, alignment, whitespace and predictable reading order.

4. **Context**
   Expose period, scope, definitions, filter state and data freshness.

5. **Story points**
   Sequence views or dashboards so each point advances the argument.

6. **Performance**
   Limit marks, heavy filters and redundant views so interaction stays responsive.

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Dashboard purpose

CONCEPT

1. One decision: A dashboard is strongest when it serves one primary decision.
2. Supporting questions: Each view answers a question needed to make that decision.

CONTROL / DECISION

1. Audience: Detail and interaction match role and literacy.
2. Success: Define what a user should know or do after using it.

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Dashboard development process

Discover

- Clarify stakeholder questions, metrics and constraints.

Sketch

- Plan hierarchy, layout and interaction before polishing.

Build

- Create validated worksheets and assemble containers.

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Information hierarchy

1

Top — Decision headline and a small KPI strip.

2

Middle — Trend, comparison and driver evidence.

3

Bottom — Detail, definitions, notes and supporting exploration.

4

Emphasis — One dominant message per screen.

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Fixed versus automatic size

1. **Fixed**
   Predictable layout for a known presentation or device size.

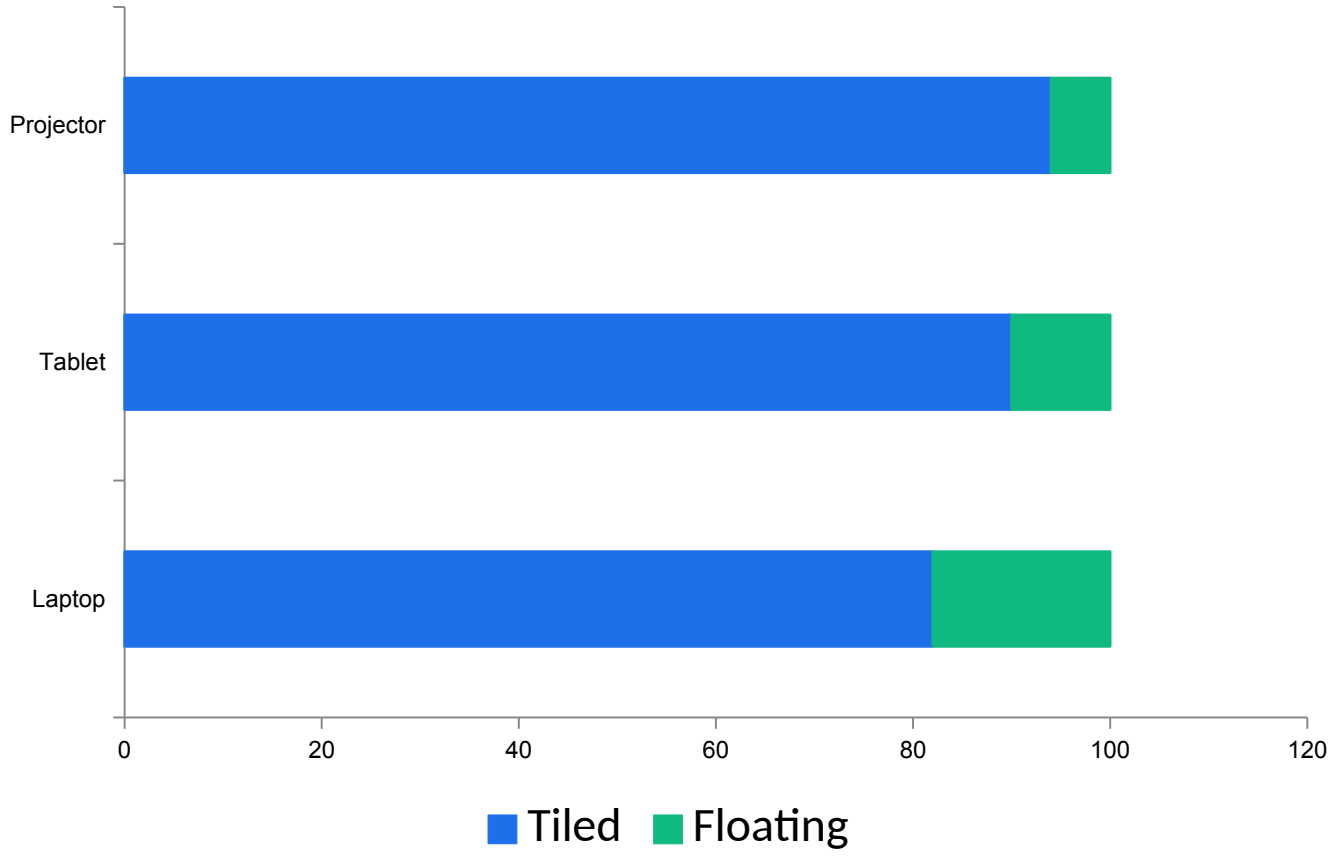
2. **Range**
   Flexible within a defined minimum and maximum.

3. **Automatic**
   Adapts to container but may reflow unpredictably.

4. **Choice**
   Design for the consumption environment, not the authoring monitor.

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Tiled and floating objects



Tiled
Reliable alignment, resizing and reading order.

Floating
Useful for deliberate overlays and precise emphasis.

Risk
Too many floating objects create collisions across sizes.

Control
Use containers and a spacing system before manual placement.

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KPI cards

CONCEPT

1. Value: Current result with unit and period.
2. Comparison: Target, prior period or benchmark.

CONTROL / DECISION

1. Direction: Change with meaningful sign and threshold.
2. Context: Definition or tooltip for ambiguous metrics.

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Filters

Relevant

- Expose only filters that change a real decision.

Scope

- Apply to selected worksheets or all using the source deliberately.

State

- Show what is selected in titles or context text.

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Use a view as a filter

1

Source — The selected mark or view controls context.

2

Target — Only relevant views update.

3

Clearing — Choose show all, exclude all or keep filtered values intentionally.

4

Test — Verify single, multi-select and clear behaviour.

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Highlight actions

1

Purpose

Emphasise related marks without removing the rest.

2

Context

Preserves the comparison base.

3

Trigger

Hover, select or menu changes the interaction feel.

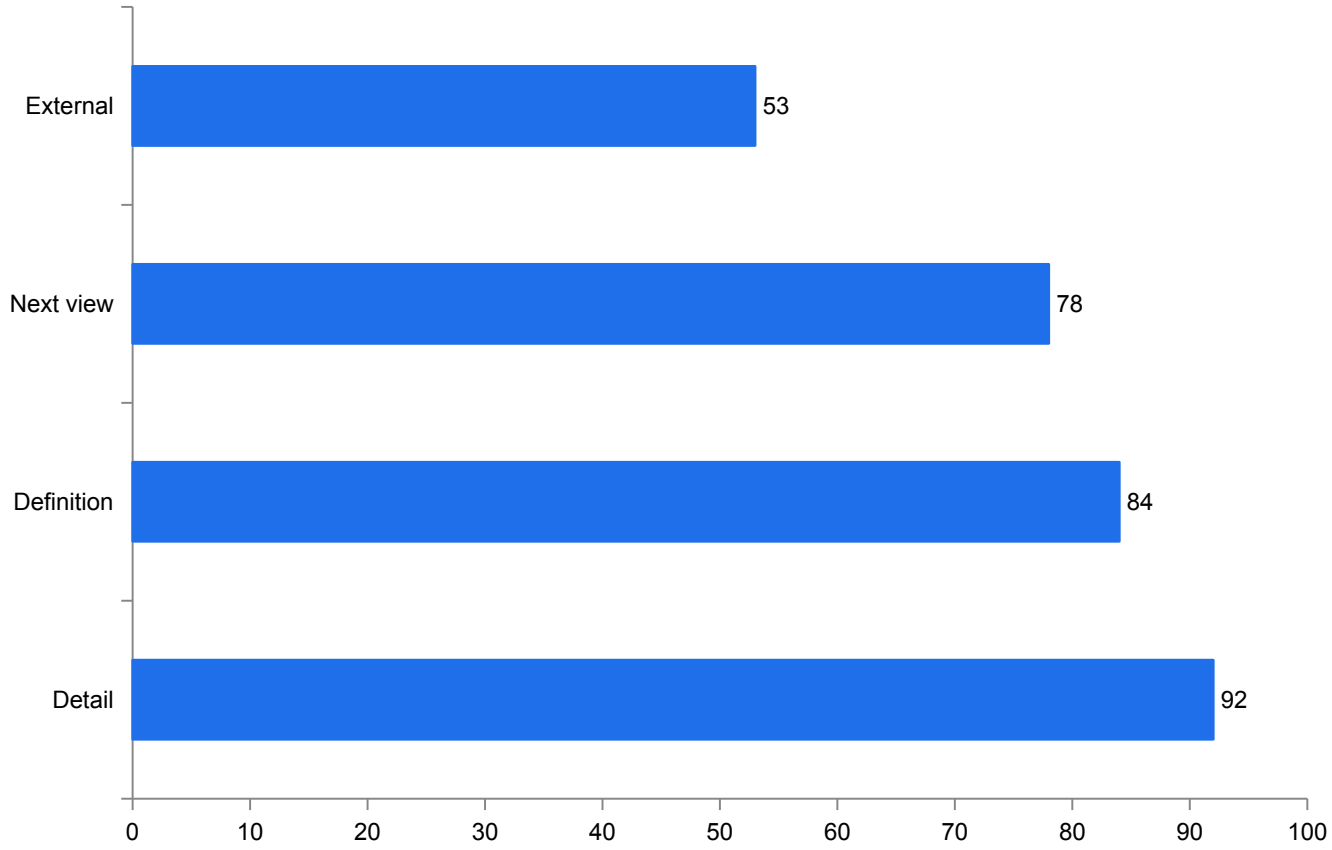
4

Use

Good for cross-view relationship and presentation dialogue.

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URL and navigation actions



URL
Open approved detail, documentation or record context.

Navigation
Move to another dashboard or story sheet.

Safety
Construct links from trusted fields and test missing values.

Affordance
Make the clickable area and destination clear.

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Tooltips

CONCEPT

1. Question: Use tooltip content to answer the likely next question.
2. Order: Lead with label and decisive value.

CONTROL / DECISION

1. Format: Keep units, decimals and dates consistent.
2. Viz in Tooltip: Use sparingly; hidden complexity can slow and distract.

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The Tableau Story workspace

Story point

- A view, dashboard or text point in a narrative sequence.

Caption

- Summarises the takeaway and advances the case.

State

- Filters and annotations can be saved per point.

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Story sequencing patterns

1

Start big — Lead with portfolio or executive context.

2

Drill down — Move from segment to driver to record evidence.

3

Highlight outlier — Explain why one exception matters.

4

Answer the question — Close with decision, owner, trigger and next evidence.

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Dashboard performance

1. **Marks**
   Reduce unnecessary marks and high-cardinality labels.

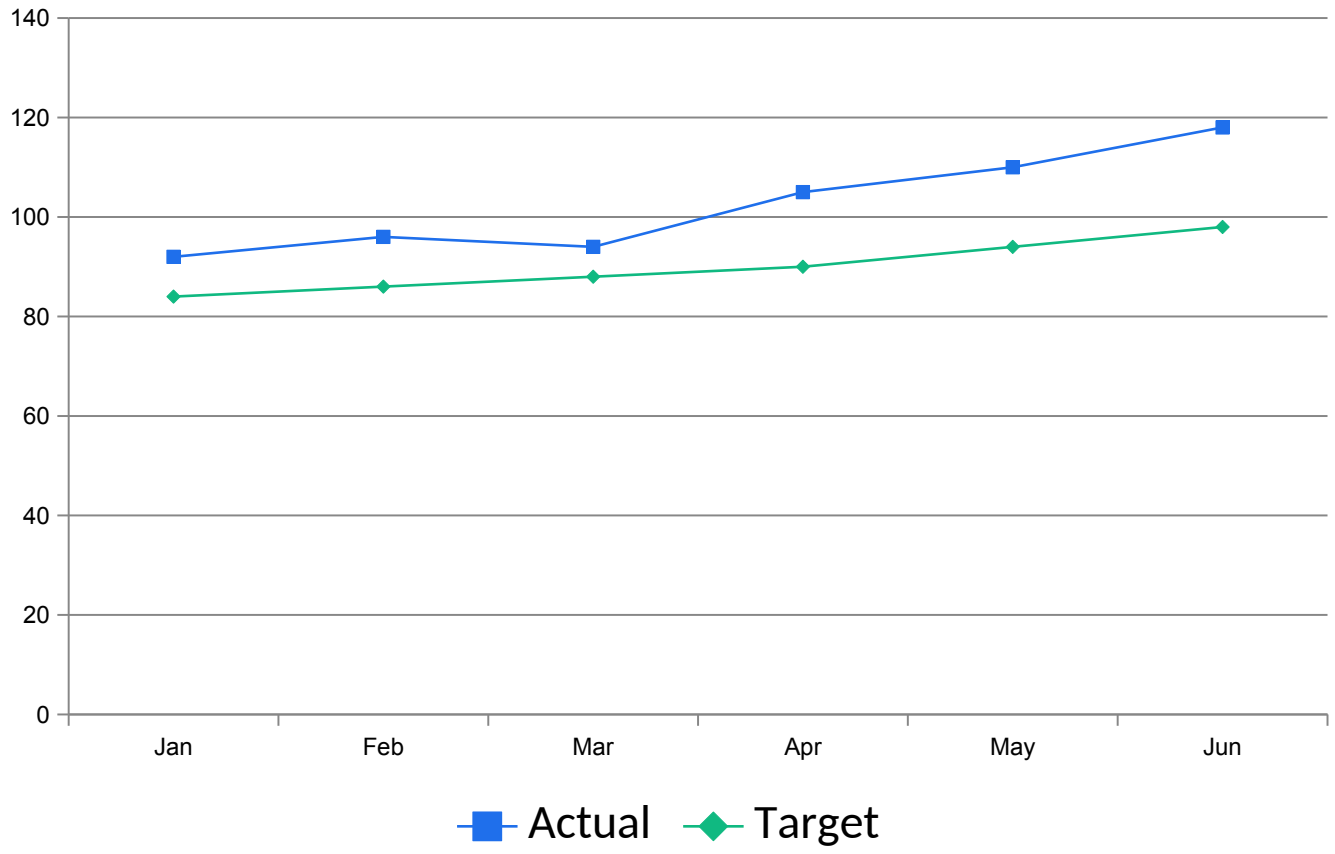
2. **Filters**
   Avoid many expensive quick filters and broad context filters.

3. **Calculations**
   Prefer efficient logic and governed aggregations.

4. **Test**
   Measure load and interaction on the actual publication environment.

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Dashboard quality gate



Numbers
Reconcile totals and definitions to source controls.

Interaction
Test source, target, trigger and clear state for every action.

Visual
Check hierarchy, clipping, contrast and reading order.

Stakeholder
Confirm the dashboard answers the key questions without analyst narration.

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TOPIC 03 · HANDS-ON

Assemble an Omnichannel Executive Dashboard

LAB 5

Build a fixed-size executive dashboard that connects KPI, trend, channel and store evidence for an omnichannel retailer.

EXPECTED OUTPUT

A 1366 x 768 dashboard with KPI strip, revenue trend, channel comparison, store drill-down and visible context.

TOOLS Tableau Public/Desktop · Excel · Dashboard containers

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LAB 5 · ACCEPTANCE

Assemble an Omnichannel Executive Dashboard

ACCEPTANCE CHECK

The dashboard answers the executive question in under 30 seconds and exposes period, filters, definitions and freshness.

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TOPIC 03 · HANDS-ON

Create a Branch Turnaround Tableau Story

LAB 6

Sequence dashboards into a Tableau Story that moves from portfolio context to branch diagnosis and an accountable turnaround decision.

EXPECTED OUTPUT

A four-point Tableau Story with captions, annotations, saved filter states and a clear call to action.

TOOLS Tableau Public/Desktop · Excel · Story workspace

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LAB 6 · ACCEPTANCE

Create a Branch Turnaround Tableau Story

ACCEPTANCE CHECK

Each story point advances the argument, the filter state is intentional, and the last point names an owner and next action.

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TOPIC RECAP

Recap — Develop Dashboards and Stories

1

Knowledge

Dashboard purpose and related principles.

2

Ability

LO3 · A3 · K3

3

Evidence

A validated workbook and decision-ready interpretation.

4

Next

Connect this topic to the next layer of the analytical workflow.

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Lunch Break

45 minutes

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04

TOPIC 04

Present and Review Stories

Stakeholder narrative · presentation · SME review · evidence traceability

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ALIGNMENT A4 · A6 · K4

Key Concepts — Present and Review Stories

1. **Executive opening**
   Lead with the decision, impact and confidence level.

2. **Evidence ladder**
   Move from KPI to trend, driver, segment and record-level detail.

3. **Interaction discipline**
   Use filters to answer likely questions without losing the narrative.

4. **SME review**
   Separate data facts, analytical interpretation and domain judgement.

5. **Revision log**
   Record what changed, why, who reviewed it and which evidence supports it.

6. **Accessibility**
   Use descriptive titles, sufficient contrast, limited marks and non-colour cues.

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Present for the decision

CONCEPT

1. Open: State the decision, impact and timing.
2. Evidence: Show only the views required to support the claim.

CONTROL / DECISION

1. Implication: Translate the pattern into stakeholder consequence.
2. Ask: Name the approval, intervention or next evidence required.

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Start big

Portfolio

- Orient the audience to scale and overall performance.

Benchmark

- Show target, peer or prior-period context.

Materiality

- Explain why the issue deserves attention.

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Drill down deliberately

1

Segment — Identify where the aggregate change originates.

2

Driver — Separate volume, rate, mix and operational mechanism.

3

Record — Inspect detail only when it changes confidence or action.

4

Return — Reconnect the detail to the original decision.

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Highlight outliers

1. **Expected**
   Define the comparison band or target.

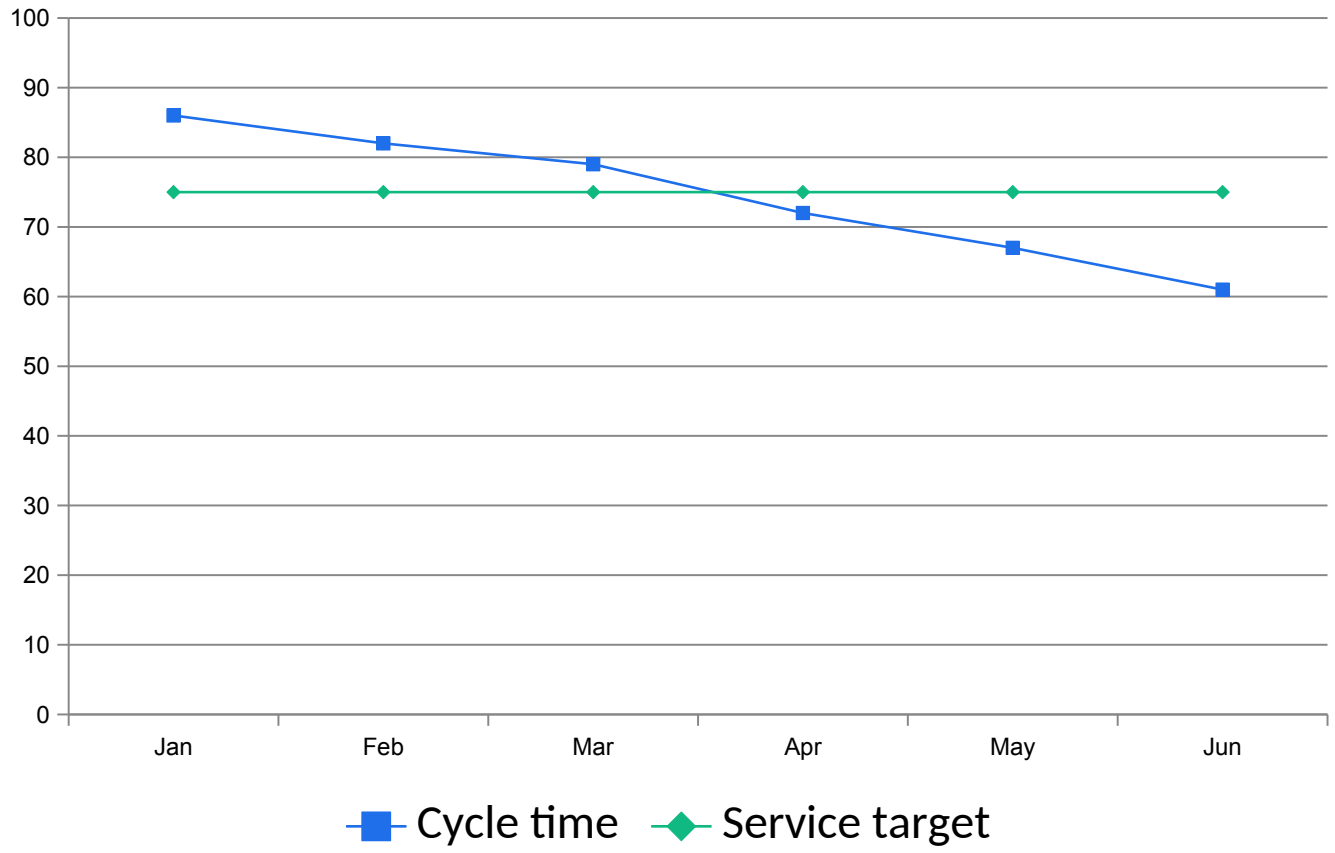
2. **Observed**
   Mark the exception clearly.

3. **Validate**
   Check source and definition before interpretation.

4. **Respond**
   Choose investigation, correction, containment or monitoring.

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Show a trend



Window
Select a time range that reveals rather than hides the change.

Granularity
Use consistent intervals and enough observations.

Annotation
Mark interventions, events and change points.

Caution
Do not imply a forecast or cause that the evidence cannot support.

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Offer analysis, not chart narration

CONCEPT

1. Weak: The blue line goes up.
2. Stronger: Conversion fell after week 14 while traffic remained stable.

CONTROL / DECISION

1. Decision: The evidence supports investigating checkout friction before buying more traffic.
2. Limitation: Campaign mix changed in the same period, so isolate it next.

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Answer the stakeholder question

Restate

- Use the stakeholder's decision language.

Resolve

- Give a direct answer with confidence.

Support

- Cite the decisive view and metric.

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Plan for live questions

1. Known — Prepare supporting views for likely objections.
2. Unknown — Say what the current evidence cannot answer.
3. Explore — Use interaction without abandoning the main thread.
4. Capture — Record unresolved questions, owner and due date.

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Role of the subject-matter expert

1. **Definition**
   Confirms what the metric means in practice.

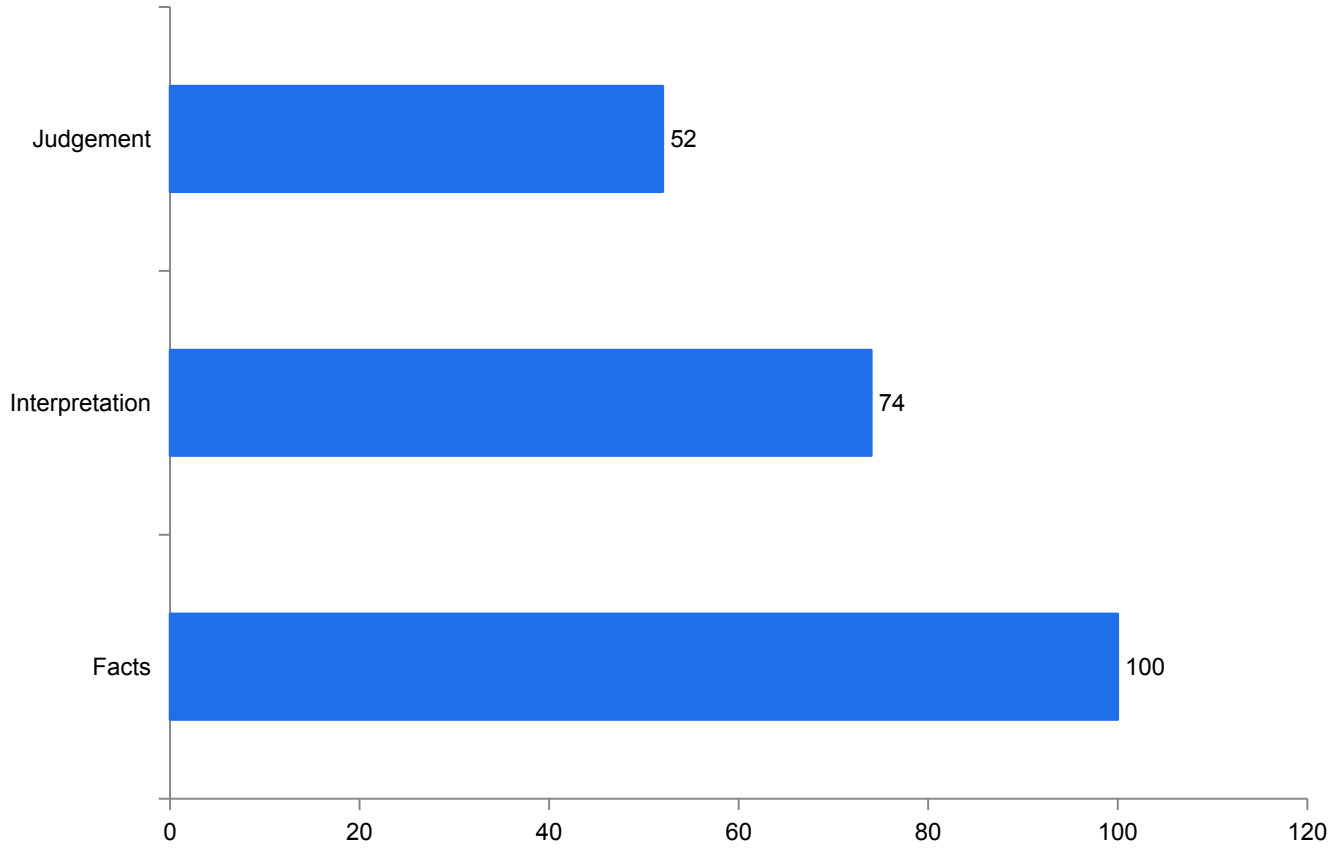
2. **Mechanism**
   Tests whether the proposed explanation is operationally plausible.

3. **Threshold**
   Defines materiality, risk and acceptable performance.

4. **Action**
   Validates feasibility and unintended consequences.

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Facts, interpretation and judgement



Fact
A value or relationship directly represented in the evidence.

Interpretation
Analytical meaning inferred from the facts.

Judgement
Domain decision about risk, action or priority.

Discipline
Label each layer so disagreement can be resolved precisely.

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SME review protocol

CONCEPT

1. Prepare: Share question, definitions, source and draft story.
2. Review: Test accuracy, completeness, materiality and actionability.

CONTROL / DECISION

1. Resolve: Document agreed change and unresolved issue.
2. Release: Record reviewer, version, date and decision impact.

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Review dynamic displays

Questions

- Do filters and actions answer actual stakeholder questions?

State

- Can the reviewer see active scope and reset it?

Consistency

- Do labels, units and definitions remain stable across views?

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Presentation mode

1

Sequence — Know the intended path before entering full screen.

2

Pointer — Direct attention to evidence, not decoration.

3

Pace — Pause for interpretation and stakeholder questions.

4

Fallback — Keep a static export or screenshot for unavailable interaction.

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Accessible presentation

1. **Language**
   Use descriptive titles and plain terms.

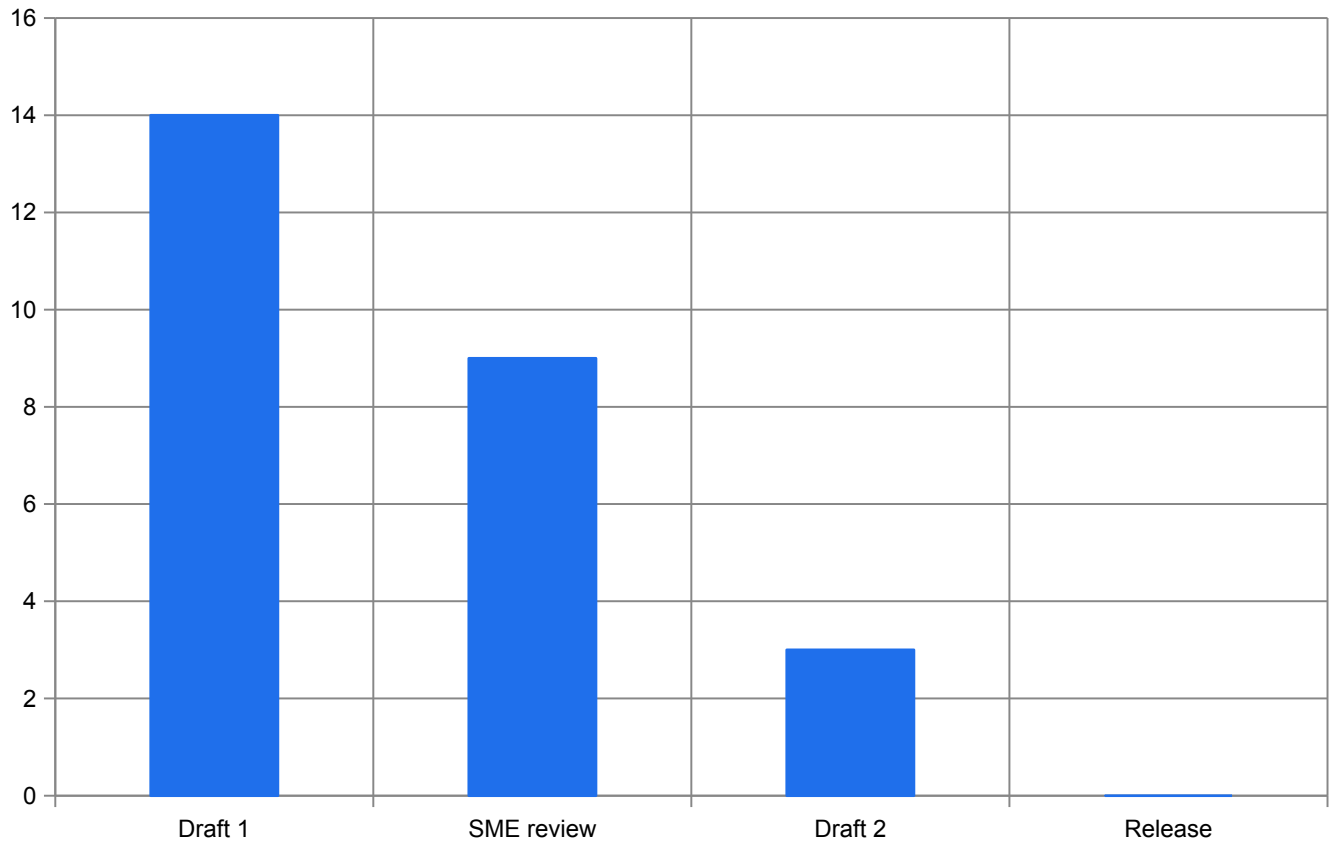
2. **Contrast**
   Ensure text, marks and controls remain legible.

3. **Redundancy**
   Do not rely on colour alone.

4. **Narration**
   Verbally explain the decisive pattern and filter state.

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Revision and release evidence



Change
What title, calculation, filter, data or conclusion changed?

Reason
Which review finding required it?

Reviewer
Who approved the definition or interpretation?

Impact
Did the recommended decision change?

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TOPIC 04 · HANDS-ON

Present a Hospital Flow Story

LAB 7

Present a patient-flow dashboard to clinical and operations stakeholders, then capture questions and decisions without over-claiming clinical causation.

EXPECTED OUTPUT

A stakeholder-ready dashboard, five-minute presentation script and question/evidence log.

TOOLS Tableau Public/Desktop · Excel · Presentation mode

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LAB 7 · ACCEPTANCE

Present a Hospital Flow Story

ACCEPTANCE CHECK

The presentation answers the operations question, identifies a limitation and records unresolved SME questions for follow-up.

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TOPIC 04 · HANDS-ON

Run an SME Review and Evidence Revision

LAB 8

Review a sustainability dashboard with finance and facilities SMEs, reconcile definition differences and release a traceable revision.

EXPECTED OUTPUT

A reviewed dashboard, metric-definition register, decision log and v2 release note.

TOOLS Tableau Public/Desktop · Excel · Review checklist

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LAB 8 · ACCEPTANCE

Run an SME Review and Evidence Revision

ACCEPTANCE CHECK

Every material claim has a definition, evidence source and reviewer; the change log distinguishes correction from interpretation.

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TOPIC RECAP

Recap — Present and Review Stories

1

Knowledge

Executive opening and related principles.

2

Ability

LO4 · A4 · K4

3

Evidence

A validated workbook and decision-ready interpretation.

4

Next

Connect this topic to the next layer of the analytical workflow.

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05

TOPIC 05

Create Interactive Stories

Calculated fields · parameters · filters · actions · navigation · user experience

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ALIGNMENT A5 · K5

Key Concepts — Create Interactive Stories

1. **Calculated fields**
   Create governed measures and business logic when source fields are insufficient.

2. **Parameters**
   Expose a controlled value that can drive calculations, filters or reference lines.

3. **Filter actions**
   Use a selected mark in one view to constrain another view.

4. **Highlight actions**
   Preserve context while emphasizing related marks.

5. **URL and navigation**
   Connect the dashboard to detail, definitions or another analytical view.

6. **Experience design**
   Make state, affordance, reset and next action visible to the user.

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Calculated fields

CONCEPT

1. Purpose: Create missing business logic from available fields.
2. Components: Functions, fields, operators, literals and parameters.

CONTROL / DECISION

1. Aggregation: Keep row-level and aggregate logic consistent.
2. Governance: Name, comment, format and test calculations.

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Types of calculation

Basic

- Row-level or aggregate expressions evaluated with the query.

Table

- Computed on the marks returned to the view.

LOD

- Controls the dimensional level used for an aggregate.

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Profit and profit ratio

1

Profit — SUM([Sales]) - SUM([Cost]).

2

Ratio — SUM([Profit]) / SUM([Sales]), not AVG of row ratios.

3

Zero — Define behaviour when the denominator is zero.

4

Format — Use currency for profit and percentage for ratio.

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Parameters

1

Value

A workbook-global placeholder selected by the user.

2

Control

Expose a list, range or all-values input.

3

Use

Drive calculations, filters, reference lines or bins.

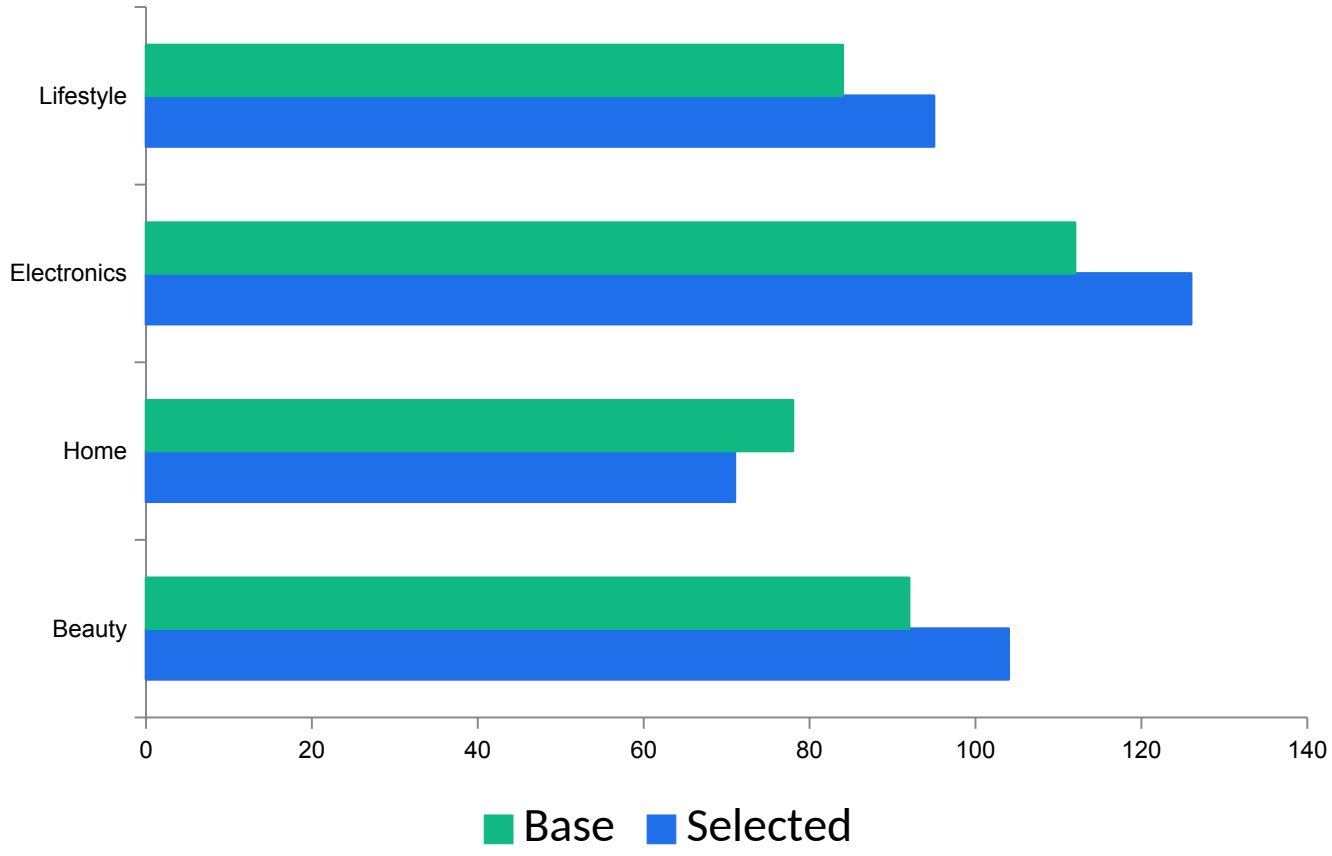
4

Caution

A parameter is not automatically connected to data until logic uses it.

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Metric selector



Parameter
Revenue, Profit or Profit Ratio.

Calculation
CASE selects the measure.

Title
Display the selected metric in the view title.

Format
Account for different units or use separate formatted views.

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Threshold explorer

CONCEPT

1. Parameter: Manager-selected discount or risk threshold.
2. Calculation: Classify marks relative to the threshold.

CONTROL / DECISION

1. Reference: Show the selected threshold visually.
2. Decision: Explain what crossing the threshold triggers.

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Filter actions

Source

- A selected mark or dashboard view.

Target

- One or more related views.

Fields

- All fields or explicit source-target mapping.

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Highlight actions

1

Preserve context — Non-selected marks remain visible.

2

Relationship — Emphasise matching category, customer or geography.

3

Trigger — Hover supports discovery; select supports deliberate focus.

4

Accessibility — Add label or shape so colour is not the only cue.

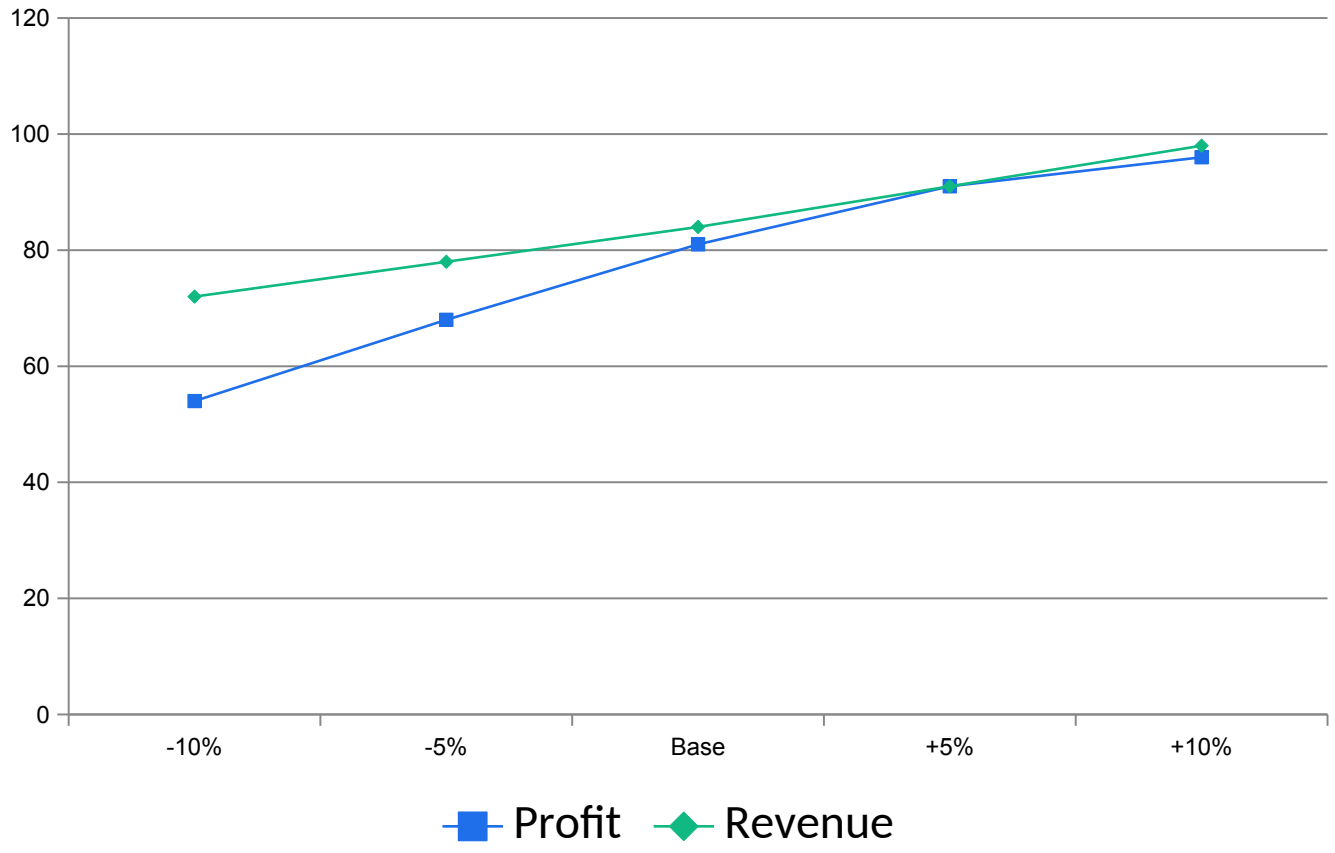
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URL actions

1. **Destination**
   Approved product page, evidence record or definition.
2. **Field insertion**
   Build the URL from encoded, trusted field values.
3. **Missing data**
   Test null and malformed values.
4. **Privacy**
   Never expose restricted identifiers through a public link.

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Parameter actions



Direct manipulation

Selecting a mark writes a value to a parameter.

Use

Reference line, selected metric, comparison target or scenario.

State

Show the parameter value visibly.

Reset

Provide a predictable default or reset control.

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Set actions

CONCEPT

1. Membership: Interaction changes which members belong to a set.
2. Use: Cohort comparison, proportional brushing and dynamic grouping.

CONTROL / DECISION

1. Context: In-set and out-of-set marks remain comparable.
2. Complexity: Explain the interaction; hidden set logic can confuse users.

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Navigation and dynamic experience

Navigation

- Move between summary and detail dashboards.

Show / hide

- Reveal optional controls or explanations without permanent clutter.

State

- Maintain clear location, selection and next action.

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Interactive mapping

1

Geography — Validate roles or use supplied latitude and longitude.

2

Layers — Separate context, performance and selection intentionally.

3

Drill — Country to region to city only when hierarchy and data grain support it.

4

Companion — Pair the map with a ranking for accurate comparison.

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Interaction quality checks

1. **Discoverability**
   Can a first-time user tell what is interactive?

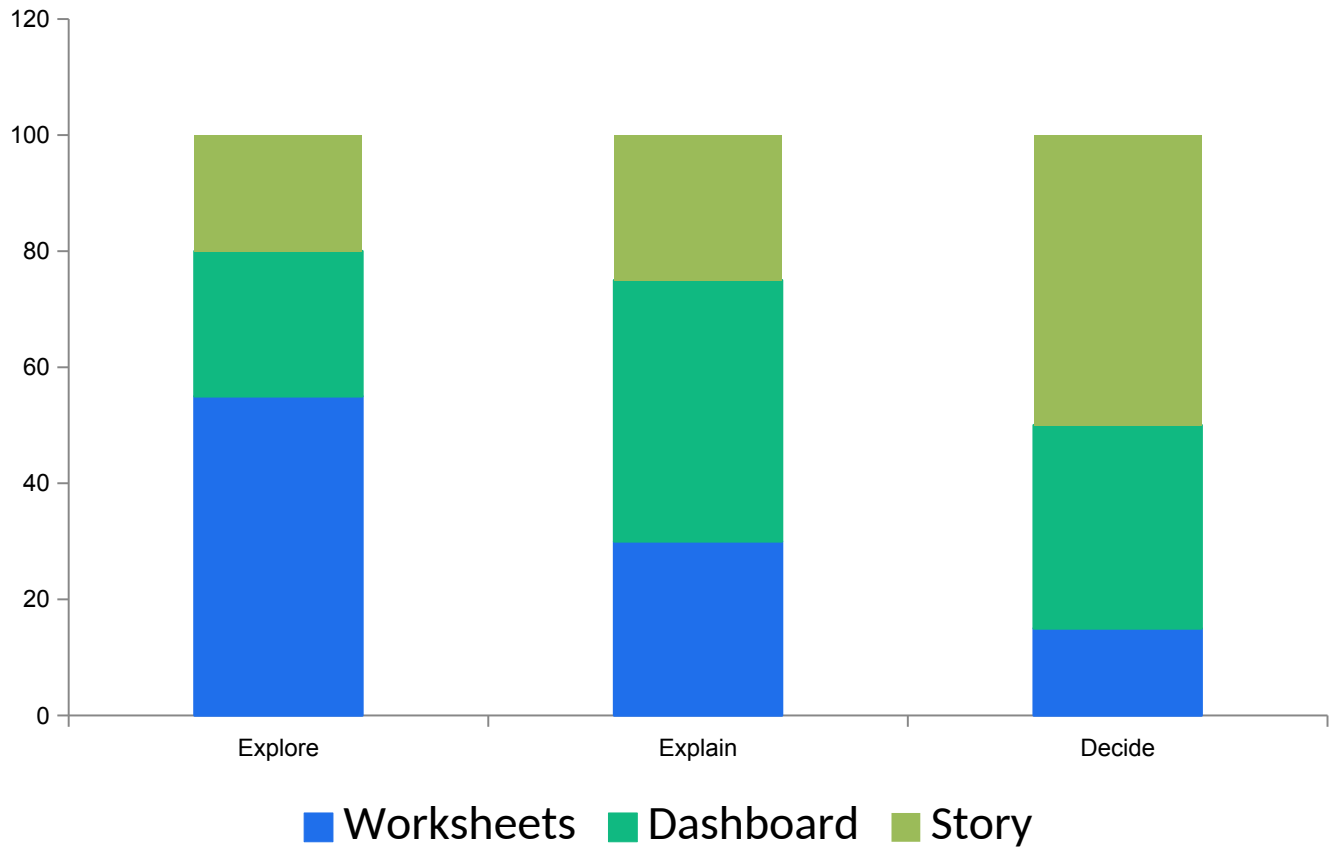
2. **Feedback**
   Does the dashboard visibly respond to the action?

3. **Recovery**
   Can the user undo or reset?

4. **Consistency**
   Do similar controls behave the same way?

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Capstone architecture



Model
Validated relationships across sales, targets and products.

Views
KPI, trend, map, category and product evidence.

Actions
Filter, highlight and navigation with visible state.

Story
Context, risk, driver and decision reviewed by an SME.

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TOPIC 05 · HANDS-ON

Build a Parameter-Driven Pricing Explorer

LAB 9

Create an interactive pricing view where managers can switch metrics, set a discount threshold and inspect product risk.

EXPECTED OUTPUT

A dashboard with metric parameter, discount threshold, dynamic title, parameter action and reset guidance.

TOOLS Tableau Public/Desktop · Excel · Parameters · Calculations

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LAB 9 · ACCEPTANCE

Build a Parameter-Driven Pricing Explorer

ACCEPTANCE CHECK

A user can switch the metric, change the threshold, select a product and reset the view without editing a worksheet.

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TOPIC 05 · HANDS-ON

Deliver an Interactive Regional Story Capstone

LAB 10

Deliver a decision-ready interactive story for a regional retailer using calculations, map, dashboard actions, story points and SME review evidence.

EXPECTED OUTPUT

A packaged Tableau workbook with executive dashboard, geographic drill-down, interactive story and signed review checklist.

TOOLS Tableau Public/Desktop · Excel · Maps · Actions · Story points

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LAB 10 · ACCEPTANCE

Deliver an Interactive Regional Story Capstone

ACCEPTANCE CHECK

The packaged workbook opens with embedded data; interactions work; the story answers the decision; and A1–A6 evidence is traceable.

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TOPIC RECAP

Recap — Create Interactive Stories

1

Knowledge

Calculated fields and related principles.

2

Ability

LO5 · A5 · K5

3

Evidence

A validated workbook and decision-ready interpretation.

4

Next

Connect this topic to the next layer of the analytical workflow.

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99

WRAP-UP

Evidence, Assessment & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1
Utilise Tableau for data visualisation.

2

LO2
Identify trends and patterns from data.

3

LO3
Develop dashboards and stories for data storytelling.

4

LO4
Present stories to stakeholders and review them with subject-matter experts.

5

LO5
Design interactive visualisations for data storytelling.

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Courseware and Assessment on the LMS

Courseware

- Learner slide PDF
- Learner Guide PDF
- Individual lab folders

Assessment

- WA question paper
- PP question paper
- Submit completed evidence

Support

- Check the current TGS code
- Use the current release
- Contact the trainer or academy

https://lms-tms.tertiaryinfotech.com/

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FINAL ASSESSMENT

Assessment Reminder

1

WA

5 open-ended knowledge questions, 1 hour.

2

PP

5 connected Tableau tasks using the supplied Excel data, 90 minutes.

3

Open book

slides, Learner Guide and approved materials.

4

Attendance

Complete assessment digital attendance before starting.

5

Submission

Upload only your own completed question papers and evidence.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then PP
4. Submit on the LMS
5. Sign Assessment Summary Record

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Digital Attendance (Mandatory)

1

Assessment attendance

Scan the SSG QR code before the papers.

2

Training attendance

Confirm AM and PM records for both days.

3

Minimum attendance

At least 75% is required.

4

Ask for help

Tell the trainer immediately if submission fails.

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THANK YOU

Turn evidence into a decision people can act on.

Frame the question, reveal the pattern, explain the implication and let interaction serve the story.

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